Data Analytics Internship Portfolio

End-to-End Analytics Journey

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Transforming Raw Data into Actionable Business Insights

About Me

Aspiring Data Analyst with hands-on experience across the full analytics lifecycle — from raw data to business recommendations.

- Data Analytics
- Business Intelligence
- Data Visualization
- Statistical Analysis
- Data Storytelling

Internship Journey at a Glance

Four projects covering the complete analytics workflow — from raw data to business decisions.

Data Wrangling

EDA & BI

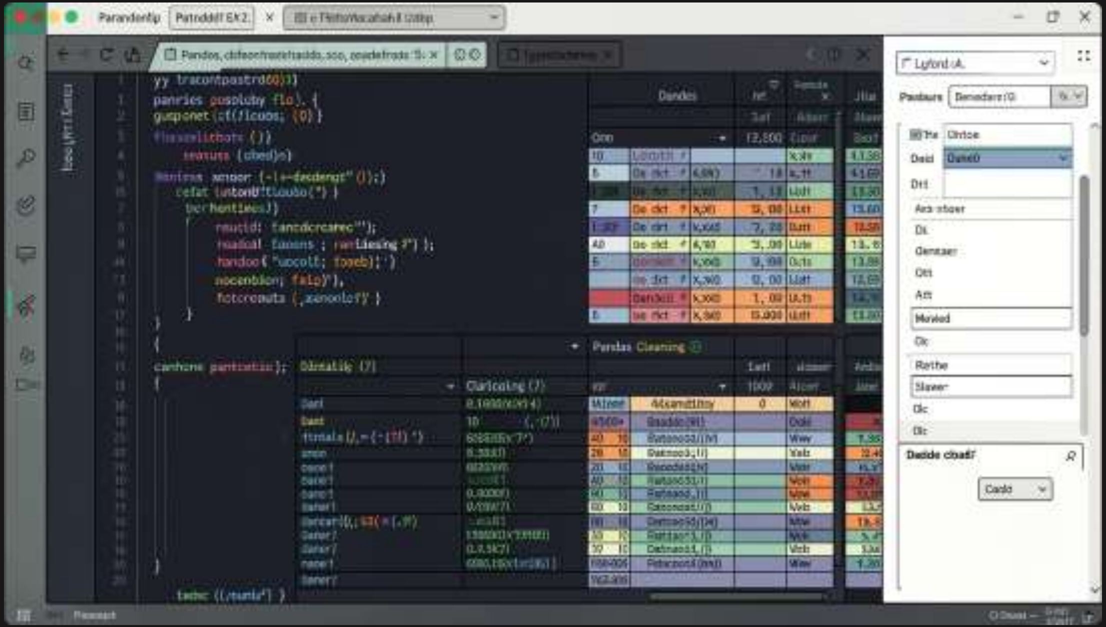
Dashboarding

Statistical Validation

Raw Data → Cleaning → Analysis → Dashboarding → Statistical Validation → Business Recommendations

TASK 1

Data Immersion & Wrangling



Objective

Prepare raw project management data for analysis.

Techniques Applied

- Missing value handling & duplicate removal
- Date standardization & feature engineering
- Task Status, Duration & Progress % fields

Tools

Python · Pandas · NumPy · Google Colab · GitHub

Data Immersion & Wrangling

Objective:

Prepare raw project management data for analysis by identifying and resolving data quality issues.

Dataset Loading

```
import pandas as pd
import numpy as np
df = pd.read_csv("project_management_dataset.csv")
df.head()
```

Purpose:

Import dataset into Google Colab

Preview records and understand structure

Begin data familiarization process

Output:

Dataset successfully loaded

Initial understanding of project data

Dataset Profiling & Quality Assessment

Understanding Dataset Structure:

df.info()
Statistical Summary
df.describe()
Dataset Dimensions
df.shape

Purpose:
Identify data types
Understand numerical distributions
Determine dataset size
Detect potential quality issues

Key Findings:
Total rows and columns identified
Numerical fields analyzed
Data quality issues discovered

Missing Value Analysis:

Check Missing Values:

```
df.isnull().sum()
```
Missing Value Percentage
```
(df.isnull().sum() / len(df)) * 100
```

Purpose:

Identify incomplete records
Determine impact on analysis
Plan cleaning strategy

Output:

Missing values detected
Columns requiring treatment identified

Data Transformation & Standardization

Date Formatting:

```
df['Start_Date'] = pd.to_datetime(df['Start_Date'])
df['End_Date'] = pd.to_datetime(df['End_Date'])
```

Standardize Text Fields

```
df['Priority'] = df['Priority'].str.upper()
```

Purpose:

Ensure consistency

Improve data quality

Enable accurate calculations

Result:

Standardized and validated dataset

Feature Engineering

Task Duration Calculation:

```
df['Task_Duration'] = (
    df['End_Date'] -
    df['Start_Date']
).dt.days
```

Progress Percentage

```
df['Progress_Percentage'] = (
    df['Completed_Tasks']
    /
    df['Total_Tasks']
) * 100
```

Task Status Creation

```
df['Task_Status'] = np.where(
    df['Progress_Percentage'] == 100,
    'Completed',
    'In Progress'
)
```

Task Status Creation:


```
df['Task_Status'] = np.where(
    df['Progress_Percentage'] == 100,
    'Completed',
    'In Progress'
)
```


Purpose:

create business-focused metrics for analysis.

New Features Created

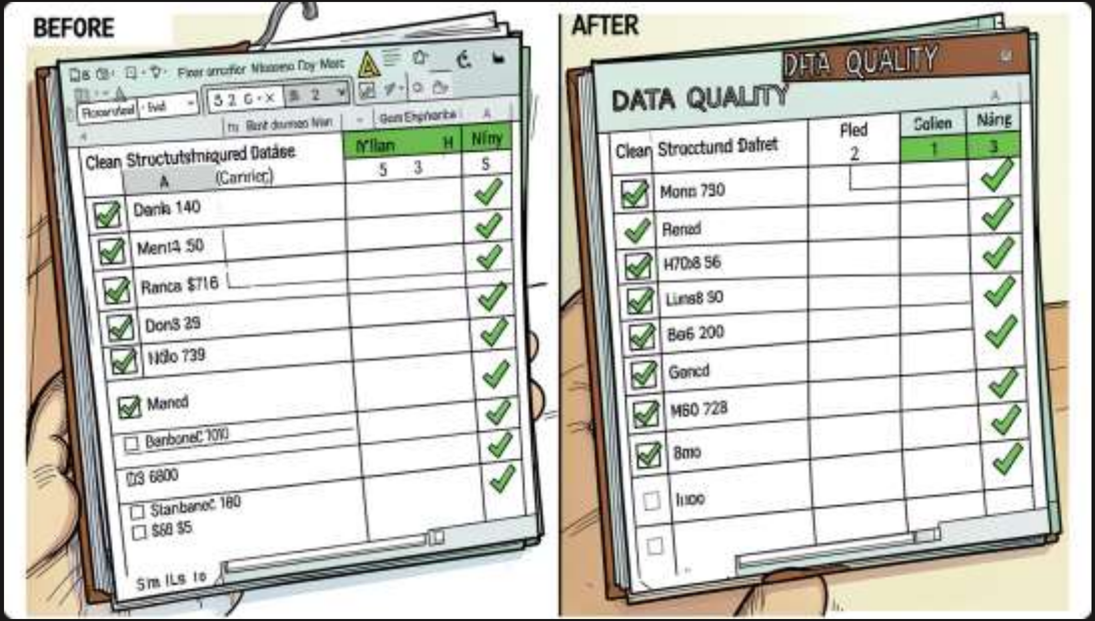
Task Duration

Progress Percentage

Task Status

TASK 1 — RESULTS

Clean Data, Reliable Insights



Outcomes

- Clean, structured, analysis-ready dataset
- Reduced redundancy, improved consistency
- Foundation for accurate reporting & forecasting

Key Learning: Insight quality depends on data quality.

TASK 2

Exploratory Data Analysis & Business Intelligence

Analyzed employee data using SQL and Python to answer key business questions.

Salary Analysis

Highest, lowest, average salaries; threshold filtering

Department Insights

Workforce distribution and salary spend by department

Employee Ranking

Ranked employees by salary metrics for compensation review

```python
from google.colab import files
uploaded = files.upload()
```

Choose Files No file chosen Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.

```
Saving Employee_source_Dataset--lab06.xlsx to Employee_source_Dataset--lab06.xlsx
```

```python
import pandas as pd

df = pd.read_excel("Employee_source_Dataset--lab06.xlsx")

df.head()
```

	Emp_ID	Name	Department	Salary
0	101	Ali	IT	120000
1	102	Sara	HR	75000
2	103	John	Finance	45000
3	104	Ayesha	IT	98000
4	105	David	Marketing	30000

```python
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10 entries, 0 to 9
Data columns (total 4 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   Emp_ID      10 non-null     int64
 1   Name        10 non-null     object
 2   Department  10 non-null     object
 3   Salary      10 non-null     int64
dtypes: int64(2), object(2)
memory usage: 452.0+ bytes
```

```
import sqlite3

conn = sqlite3.connect("employee.db")
```

```
df.to_sql(
    "employees",
    conn,
    if_exists="replace",
    index=False
)
```

```
10
```

```
pd.read_sql(
```

Made with GAMMA

https://colab.research.google.com/drive/1T3qyL9UL8Hwt91wVl8zPlt0trod8Tjl

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File Edit View Insert Runtime Tools Help

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Table of contents

+ Section

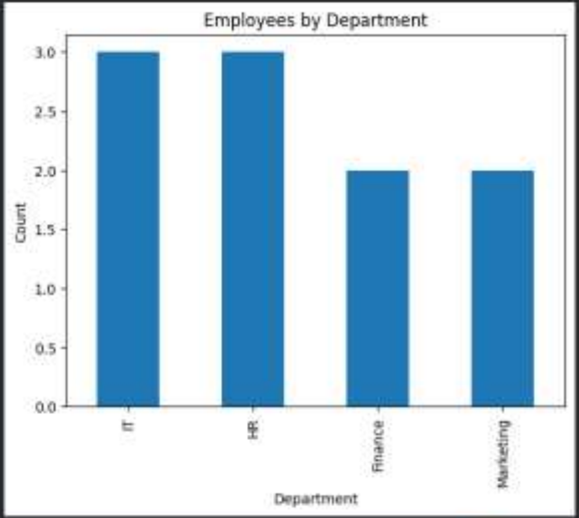
```
pd.read_sql(
    "SELECT * FROM employees",
    conn
)
```

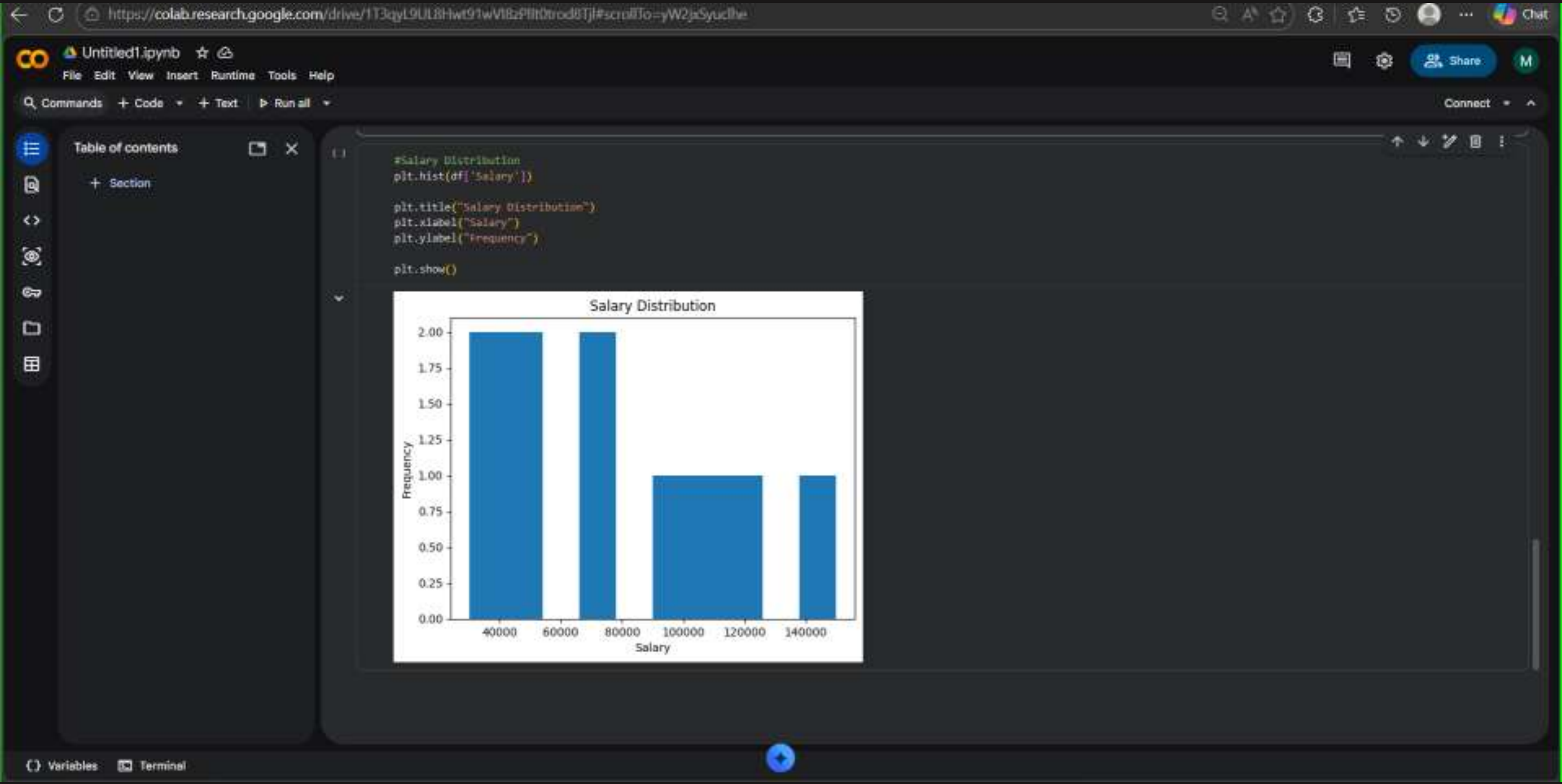
	Emp_ID	Name	Department	Salary
0	101	Ali	IT	120000
1	102	Sara	HR	75000
2	103	John	Finance	45000
3	104	Ayesha	IT	98000
4	105	David	Marketing	30000
5	106	Fatima	HR	150000
6	107	Rahul	Finance	67000
7	108	Zara	IT	52000
8	109	Omar	Marketing	40000
9	110	Neha	HR	110000

```
#Create Visualizations
#Department-wise Employee Count
import matplotlib.pyplot as plt
```

Variables Terminal

```python
import matplotlib.pyplot as plt

dept = df['Department'].value_counts()

dept.plot(kind='bar')

plt.title("Employees by Department")
plt.xlabel("Department")
plt.ylabel("Count")

plt.show()
```




Untitled1.ipynb
File Edit View Insert Runtime Tools Help
Commands
+ Code
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Connect
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Table of contents
+ Section
#Salary Distribution
plt.hist(df['Salary'])
plt.title("Salary Distribution")
plt.xlabel("Salary")
plt.ylabel("Frequency")
plt.show()
Salary Distribution
Frequency
Salary
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00
40000
60000
80000
100000
120000
140000
Variables
Terminal

TASK 3

Interactive HR Analytics Dashboard

Dashboard Components

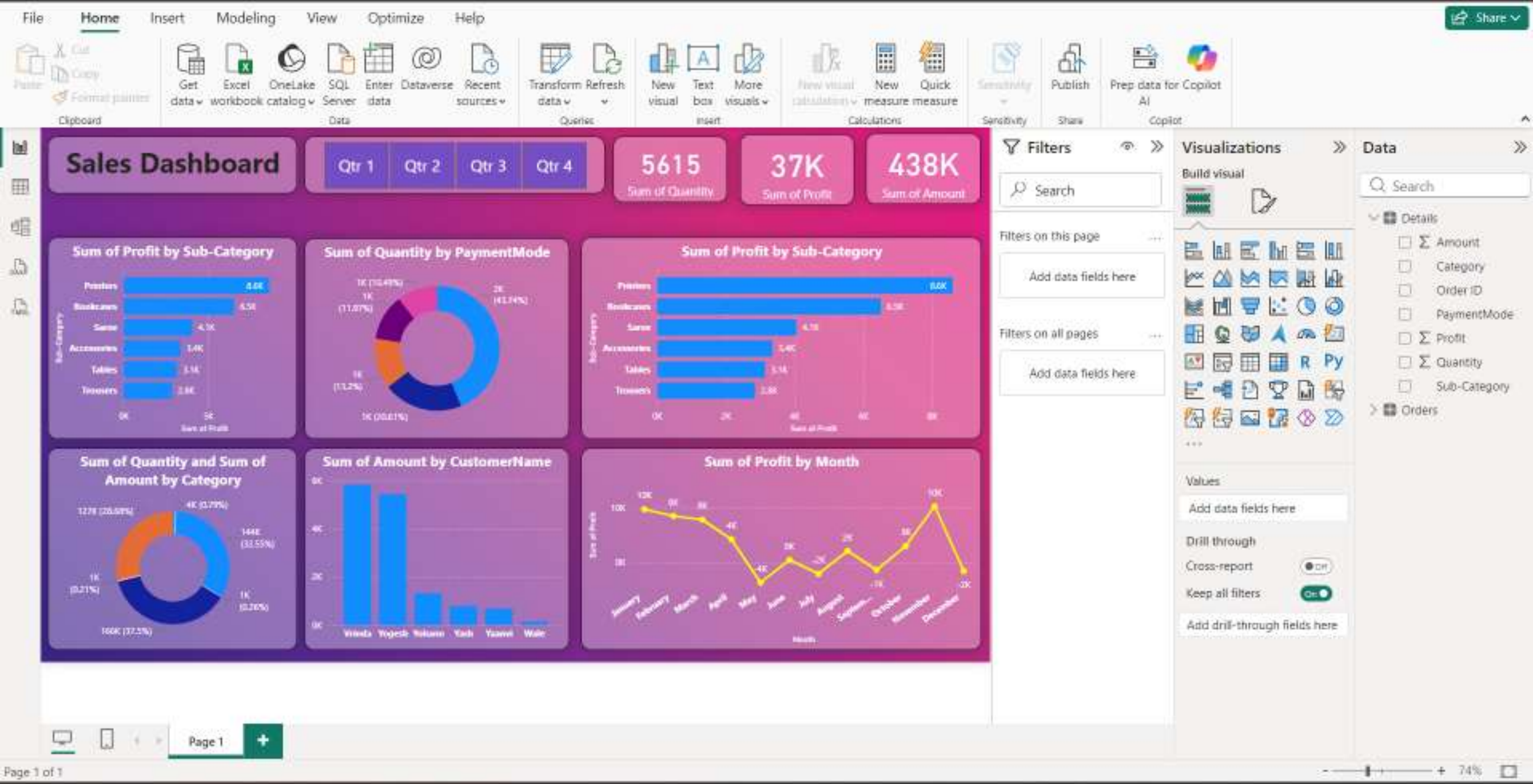
- KPIs: Total Employees, Salary Cost, Avg/Max/Min Salary
- Employee & salary distribution charts
- Interactive filters by department

Business Value

Real-time workforce monitoring, faster decisions, improved reporting efficiency.

Tools

Power BI · DAX · Excel


File
Home
Insert
Modeling
View
Optimize
Help
Share
Get data
Excel workbook
OneLake catalog
SQL Server
Enter data
Dataverse
Recent sources
Transform data
Refresh
New visual
Text box
More visuals
New measure
Quick measure
Publish
Prep data for AI
Copilot
Clipboard
Data
Queries
Insert
Calculations
Sensitivity
Share
Copilot
Sales Dashboard
Qtr 1
Qtr 2
Qtr 3
Qtr 4
5615
Sum of Quantity
37K
Sum of Profit
438K
Sum of Amount
Sum of Profit by Sub-Category
Sum of Quantity by PaymentMode
Sum of Profit by Sub-Category
Sum of Quantity and Sum of Amount by Category
Sum of Amount by CustomerName
Sum of Profit by Month
Filters
Search
Filters on this page
Add data fields here
Filters on all pages
Add data fields here
Visualizations
Build visual
Values
Add data fields here
Drill through
Cross-report
Keep all filters
Add drill-through fields here
Data
Search
Details
Amount
Category
Order ID
PaymentMode
Profit
Quantity
Sub-Category
Orders
Page 1
Page 1 of 1
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TASK 4

Business Narrative & Statistical Validation

Objective

Convert findings into a compelling, statistically validated business story.

Methods Used

- T-Test & Chi-Square Test
- P-Value analysis & confidence intervals
- Data-backed recommendations

Impact

Statistical evidence ensures conclusions are reliable — not coincidental.

Skills Acquired

Python

Pandas · NumPy

SQL

SQLite · Queries

Power BI

DAX · Dashboards

Statistics

T-Test · Hypothesis

Data Storytelling

EDA · Visualization

Key Learnings & Conclusion

Data Quality First

Clean data drives reliable insights

End-to-End Workflow

Wrangling → EDA → Dashboards → Validation

Business Communication

Translate analysis into clear recommendations

> Data becomes valuable when transformed into meaningful insights that drive better decisions.

Thank You — Kovvuri Gnana Swarupa